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Polycaprolactone Critical Gels

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ABSTRACT: The molecular weight dependence of critical gel properties was determined for poly(ϵ -caprolactone) diol end-linked with a three-functional isocyanate. The critical gels exhibit the typical scaling behavior with a power law relaxation spectrum $H(\lambda) = G_0/\Gamma(n)(t/\lambda_0)^{-n}$, $\lambda > \lambda_0$, where $G_0 = G_e$ was found to be the modulus of the fully cross-linked polymer and $\lambda_0 = \eta_0/G_e$ was found to depend on the viscosity of the difunctional prepolymer, η_0 . The relaxation exponent, n , decreases with increasing cross-linker concentration (increasing stoichiometric ratio, $r = [\text{NCO}]/[\text{OH}]$) and increasing molecular weight of the prepolymer. This suggests that the fractal dimension of the critical gel increases with increasing molecular weight and stoichiometric ratio.

Introduction

The mechanical properties of polymeric materials are highly dependent on molecular weight. The zero-shear viscosity (η_0) of a large number of polymeric liquids was found to depend on the weight-average molecular weight (M_w) as

$$\eta_0 \propto M_w^\alpha \quad (1)$$

with $1 \leq \alpha \leq 2.5$ for $M_w \leq M_e$ and $\alpha = 3.4$ for $M_w \geq M_e$, where M_e is the entanglement molecular weight.^{1,2} The equilibrium modulus (G_e) of a rubbery solid decreases with the molecular weight between cross-linking points, M_c ,³

$$G_e \propto 1/M_c \quad (2)$$

Although these relationships are well understood for polymeric liquids and solids, little is known about the molecular weight dependence of the mechanical properties of a cross-linking polymer which undergoes a transition from a liquid to a rubbery solid. The average molecular weight increases and the molecular weight distribution broadens with increasing extent of reaction, and both diverge at the gel point. The molecular weight of end-linking polymers before cross-linking determines the strand lengths in the evolving network. It, therefore, is assumed that the molecular weight of the initial polymer components (M_n) is an important parameter for the gel behavior.

Recently, rheological experiments have revealed that the critical gel (CG), which is the ideal material exactly at the gel point (GP), exhibits a continuous power law relaxation time spectrum (CW spectrum)^{4,5}

$$H(\lambda) = H_0 \lambda^{-n} \quad \lambda \leq \lambda_0 \quad (3)$$

behavior is also apparent in dynamic mechanical experiments where the storage modulus (G') and the loss modulus (G'') at GP are

$$G' = G''/\tan \delta = S \omega^n \Gamma(1-n) \cos \delta, \quad 0 < \omega < 1/\lambda_0 \quad (5)$$

The phase angle (δ) between stress and strain is independent of frequency (ω) but proportional to the relaxation exponent

$$\delta = n\pi/2 \quad (6)$$

The initial experimental studies⁴⁻⁶ for the determination of S and n focused on a cross-linking poly(dimethylsiloxane) (PDMS) system, and the effects of stoichiometry and dilution have been investigated in great detail. The value of n varied over about the entire range between 0 and 1, depending on molecular composition and cross-linking conditions.

Theories⁷⁻¹⁰ predict the power law behavior of CGs as well as the exponent (n) in conjunction with a fractal dimension which derives from the statistical self-similarity. Some theories actually suggest that the exponent (n) should be independent of molecular structure.

The complicated dimensions of S , Pa s^n , suggest that it is composed of a material characteristic modulus and a time.⁴ Experiments can be modeled with a relation between S and n of the form⁶

$$S = G_0 \lambda_0^n \quad (7)$$

where λ_0 and G_0 are material-specific constants. It was found that G_0 and $G_0 \lambda_0$ were very close to the modulus of the fully cross-linked material and the zero-shear viscosity of the prepolymer, respectively.

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